

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T) Semester – IV Examination – JUNE 2011 PT 208 BIOSTATISTICS (PART-I)

Date: 16.06.11, Thursday

Time: 1.30p.m to 1.40 p.m.

Maximum Marks: 5

Instructions:

1. There are two parts in the paper. **PART – I** contains 5 Questions and **PART – II** contains 2 Questions.
2. **PART – I** should be answered in MCQ'S question paper & **PART – II** should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 10 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 10 min can collect main question paper for **PART – II** immediately after handing over the **PART-I** question paper.

Answer the following questions

05

1. Classify the following measures used by physical therapist in clinical setting as Quantitative or Qualitative
 - (i) A muscle group is either spastic or nonspastic
 - (ii) Number of steps can a patient ambulate
 - (iii) Measurement of muscle circumference with tape
 - (iv) A patient demonstrates either normal or abnormal lumbar extension
2. A correlation equal to zero indicates that there is no relationship exists between variables.
☐ True ☐ False
3. If the null hypothesis is true and rejected, then the error committed is refer to as
☐ Type I Error ☐ Type II Error
4. The distances (meters) in six – minute walk test performed by four patients with cardiovascular disease are 635, 610, 625 and 630.
The mean distance is 625
☐ True ☐ False
5. A frequency distribution of continuous variable can be represented graphically by
☐ Scatter Diagram ☐ Dot Diagram ☐ Histogram ☐ None of These

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY	
CHARUSAT CAMPUS CHANGA	
Examination Semester	_____
Subject	_____ No. _____
Candidate's Seat No.	_____
Section	_____
Date	_____
Sr. Supervisor's Sign.	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T) Semester – IV Examination – JUNE 2011 PT 208 BIOSTATISTICS (PART-II)

Date: 16.06.11, Thursday

Time: 1:40 p.m. to 3:00 p.m.

Maximum Marks: 30

Instructions:

1. There are two parts in the paper. **PART – I** contains 5 Questions and **PART – II** contains 2 Questions.
2. **PART – I** should be answered in MCQ'S question paper & **PART – II** should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 10 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 10 min can collect main question paper for **PART – II** immediately after handing over the **PART-I** question paper.

- Q 1** Manual muscle testing with the examiner providing the resistance has long been a standard test of muscle strength. Through the use of extremities acting as levers, clinicians have been able to effectively apply resistance to all muscle groups except the ankle plantar flexors. Two hundred three subjects were studied for their ability to do standing heel-rises, as is done when testing plantar-flexion strength using the upright test. There were 122 male subjects and 81 female subjects, ranging in age from 20 to 59 years. A parallelogram electrogoniometer fitted with a digital readout was used for measuring range of motion at the ankle during the heel-rises. The following table represents the mean plantar – flexion angle (degree) in 28 repetitions for 30 male subjects. **10**

22.2	26.4	16.7	33.7	31.2	24.4
46.5	23.6	13.7	27.4	27.5	21.9
35.9	30.6	41.1	35.3	17.2	14.7
21.9	12.3	29.3	27	29.6	38.2
27.3	22.1	30.9	23.1	33	21.5

- (a) Construct Steam and Leaf table
- (b) Compute median of mean plantar- flexion angle

OR

- Q 1** For persons under the age of 45 years, low back pain (LBP) is the most common cause of activity limitation. Prolapsed disk is one of most frequent conditions. Low back pain is not a diagnosis. It is a symptom that occurs in a wide variety of medical, musculoskeletal, and neurological conditions. The specific cause of LBP is difficult to establish in most patients at the time of the initial episode. A sample of eligible 100 consecutive patients with complaints of LBP who were seen at several university – based outpatient orthopedic spine clinics were recruited to participate in this study. **10**
- The following table represents the screening test result for the LBP diagnosis type disk disease.

	Subjects with disk disease	Subject without disk disease	Total
Subject who test positive	19	62	81
Subject who test negative	6	13	19
Total	25	75	100

Using the data in above table compute

- (a) Sensitivity of test
- (b) Specificity of test
- (c) Positive Predictive value
- (d) Negative Predictive value

Q 2 Answer the following questions (Any Four)

20

- 1 Many patients with hemiplegia have good trunk balance, but others have a tendency to deviate from the normal symmetrical trunk posture in sitting. The purpose of study was to describe the incidence of sitting imbalance and its relationship to the side of predominant weakness in 115 hemiparetic patients.

The following table shows side of Hemiparesis and Sitting Balance in 115 Patients with Hemiparesis

Sitting balance	Side of Hemiparesis	
	Left	Right
Independent*	35	50
Not Independent	17	13

*Independent sitting balance was tested at the initial evaluation and defined as the ability to sit on the edge of a mat without back support, without the use of the upper extremities, without the help of an assistant, and without weight bearing through the feet.

Do the data indicate a significant relationship between the side of weakness and independence in sitting balance? Use 5% level of significance. $\chi^2_{0.05,1} = 3.841$, $\chi^2_{0.01,1} = 6.635$

- 2 A random sample of 20 tablets from a batch gives a mean ingredient content 42 mg. and standard deviation of 6 mg. Test the hypothesis that the population mean ingredient content is 44 mg. Use 5% level of significance. Table value $t_{19, 0.05} = 2.093$ (for two tail test). $t_{19, 0.05} = 1.729$ (for one tail test)
- 3 The Step Test (ST) is a measure of dynamic standing balance and paretic – lower – extremity motor control in patients with stroke. (The ST assesses an individual's ability to place one foot onto a 7.5 – cm- high step and then back down to the floor repeatedly as fast as possible for 15 seconds). In an experiment the ST score of patients is distributed according to Normal with mean 3 and standard deviation 3.25. What percentage of patients have score grater than 3.5?
 $P(Z \leq 0.154) = 0.5612$
- 4 An article by Tuzson et al in Archives of Physical Medicine and Rehabilitation reported the following data on peak knee velocity in walking (measured in degreed per second) at flexion and extension for 18 subjects with cerebral palsy.

Flexion ($^{\circ}$ /sec)	100	150	210	255	200	185	440	110	400
Extension ($^{\circ}$ /sec)	100	150	180	165	210	155	440	180	400
Flexion ($^{\circ}$ /sec)	160	150	425	375	400	400	300	300	320
Extension ($^{\circ}$ /sec)	140	250	275	340	400	450	300	300	275

- (a) Construct a scatter diagram and write your comments on relationship between flexion knee velocity and extension knee velocity.
- (b) Compute the sample correlation coefficient between flexion knee velocity and extension knee velocity.
- 5 Write a short note on One Way Analysis of Variance
